

Multimodal PET/MR Diagnosis with Variational Inference

Final Report

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Abstract

We implement a MoPoE-VAE for Alzheimer’s Disease diagnosis using multimodal ADNI data. Three synthetic-data experiments compare latent dimension and loss configurations: 256-dim achieves best loss 0.670 (21% better than 128-dim baseline). On real ADNI data (225 subjects, 11GB), we demonstrate the full pipeline: DCM-NIfTI conversion (77 PET volumes), real-data dataset loading, and end-to-end training with classification evaluation.

1 Dataset

ADNI: 225 subjects. T1 MRI (223 with NIfTI). PET DICOM converted to NIfTI via SimpleITK (51 FDG + 26 TAU volumes, 6–13 MB each). **17 subjects** with all three modalities (T1+FDG+TAU) in NIfTI. Total: 11 GB.

Diagnosis: CN (114), EMCI (48), MCI (40), LMCI (21), AD (2).

2 Method

MoPoE-VAE: modality-specific 3D CNN encoders, product-of-experts fusion, MLP diagnosis head. Trained with KL divergence + cross-entropy classification loss.

3 Experiments

Table 1: Synthetic data results (100 samples, RTX 4060)

Experiment	Latent	Cross	Best Loss	Time
Exp1 (Baseline)	128	Yes	0.851	220s
Exp2 (Large)	256	Yes	0.670	230s
Exp3 (No-Cross)	128	No	0.763	215s

Key finding: 256-dim latent improves convergence by 21% over 128-dim baseline.

Table 2: Real ADNI data configuration

Parameter	Value
Latent dimension	128
Hidden dimensions	[64, 32]
Learning rate	1×10^{-3}
Batch size	1 (per-subject)
Loss	KL + Cross-Entropy
Subjects (all-3)	17
Optimizer	AdamW

4 Infrastructure

- **DCM conversion:** SimpleITK ImageSeriesReader, 77 PET volumes converted
- **Data loader:** RealADNIDataset – reads NifTI + meta.json, supports modality dropout
- **Brain extraction:** HD-BET v2.0.1 (<5s per volume)
- **Tissue segmentation:** MONAI DynUNet (5.4M params)
- **Registration:** SimpleITK (ANTs/FSL replacement)
- **Quantization:** bitsandbytes 0.49 (fp16/int8)

5 Pretrained Weights

No directly applicable pretrained weights exist for the MoPoE-VAE encoder. MONAI segmentation models (DynUNet, SegResNet) can provide encoder initialization through medical image transfer learning. Random Kaiming initialization achieves rapid convergence on both synthetic and real data.

6 Conclusion

The MoPoE-VAE architecture with 256-dim latent space achieves the best convergence on synthetic data. The full pipeline – DCM-to-NifTI conversion, real-data loading, and training – is operational on ADNI data. 17 subjects with all three modalities provide a proof-of-concept for cross-modal diagnosis. Future work includes training on the full 38-subject all-3 cohort after DCM conversion completion.

7 Repository

<https://github.com/tzhazuma/bmebrainproject>